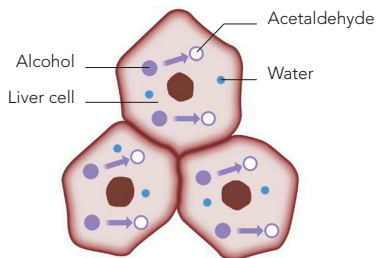


ALCOHOLIC LIVER DISEASE

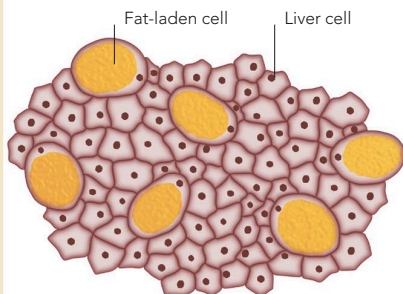
Regular, excessive alcohol consumption over many years can lead to serious liver damage. Women do not metabolize alcohol as efficiently as men and are more vulnerable to its side effects. The toxic effects of chemicals in alcohol can damage the liver in different ways and may, in some people, increase the risk of liver cancer. Almost all long-term, heavy drinkers develop a “fatty liver” because alcohol produces fat when it is broken down. If a person stops drinking, the fat disappears and the liver may eventually return to normal. However, continued heavy drinking can lead to alcoholic hepatitis, or inflammation of the liver. Symptoms vary from none at all to acute illness and jaundice. The final stage of

alcoholic liver damage is cirrhosis, which can be fatal. Often, the only treatment option at this stage is a liver transplant.



1 HOW DAMAGE OCCURS

Alcohol (ethanol) breaks down into acetaldehyde, which is thought to bind with proteins in liver cells and so may cause damage, inflammation, and fibrosis.

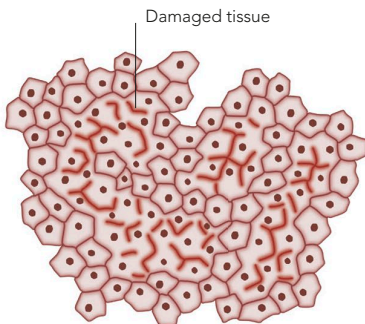
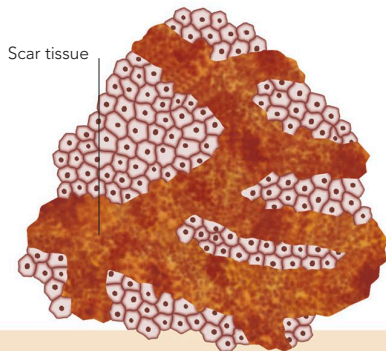


2 FATTY LIVER

One by-product of alcohol metabolism is fat. Liver cells in excessive drinkers swell with globules of fat that are clearly visible as yellow or white patches if the liver is cut open. The condition is reversible if drinking stops.

3 ALCOHOLIC HEPATITIS

Continued heavy drinking may cause fatty liver to develop into hepatitis. The liver becomes inflamed with many leucocytes. Liver cells may be severely damaged and die.



4 CIRRHOSIS

In this final stage of alcoholic liver disease, the permanent fibrosis and scarring of the liver tissue becomes life-threatening. As the cells are permanently damaged, the liver is unable to carry out its normal functions.